AI and Ethics (2024) 4:299–314
https://doi.org/10.1007/s43681-023-00266-9

ORIGINAL RESEARCH

A seven-layer model with checklists for standardising fairness assessment throughout the AI lifecycle

Avinash Agarwal[1] · **Harsh Agarwal**[2]

Received: 16 October 2022 / Accepted: 7 February 2023 / Published online: 21 February 2023

Abstract
Problem statement: Standardisation of AI fairness rules and benchmarks is challenging because AI fairness and other ethical requirements depend on multiple factors, such as context, usage case, type of the AI system, and so on. In this paper, we elaborate that the AI system is prone to biases at every stage of its lifecycle, from inception to its use, and that all stages require due attention for mitigating AI bias. We need a standardised approach to handle AI fairness at every stage. Gap analysis: While AI fairness is a hot research topic, a holistic strategy for AI fairness is generally missing. The AI fairness assessment is often developer-centric. Standard processes are required for independent audits for the benefit of other stakeholders. Also, most researchers focus only on a few facets of AI model-building. Peer review shows a focus on biases in the datasets, fairness metrics, and algorithmic bias. In the process, other aspects affecting AI fairness get ignored. The solution proposed: We propose a comprehensive approach in the form of a novel seven-layer model, inspired by the Open System Interconnection (OSI) model, to standardise AI fairness handling. Despite the differences in the various aspects, most AI systems have similar model-building stages. The proposed model splits the AI system lifecycle into seven abstraction layers, each corresponding to a well-defined AI model-building or usage stage. We also provide checklists for each layer and deliberate on potential sources of bias in each layer and their mitigation methodologies. This work will facilitate layer-wise standardisation of AI fairness rules and benchmarking parameters.

Keywords Artificial intelligence · Audit · Ethics · Fairness · Standardisation · Checklists · AI lifecycle · Trustworthy AI

1 Introduction

The use of Artificial intelligence (AI) is widespread, and AI has the potential to greatly benefit society. As we increasingly use AI systems to make critical decisions that affect people's lives, it is essential to ensure that these systems are fair and unbiased. However, the application of AI in various domains has raised concerns about the fairness of these systems. Studies demonstrate various causes of bias and how bias manifests in data [1]. Bias in AI systems can lead to negative outcomes, such as discrimination, and can undermine trust in the technology (e.g.[2, 3]). Algorithms trained on biased data can perpetuate and even exacerbate existing societal inequalities, as discussed in [4, 5].

Therefore, bias mitigation and AI fairness are becoming ethical, social, commercial, and legal requirements. Previous research has demonstrated the technical, legal, social, and ethical dimensions of bias and discrimination in AI [6], the need for regulation of AI for fairness in various domains (e.g. [7]), the need for AI fairness for people with disabilities (e.g. [8]), ethical principles for AI fairness (e.g. [9, 10]), and so on.

To address this issue, researchers have proposed various methods for assessing and addressing bias in AI systems [11]. We review some such works in the next section. However, these methods are often ad-hoc and lack a clear structure for assessing fairness at different stages of the AI lifecycle. Moreover, there is no standard process for assessing the fairness of AI systems that can be used across different domains and organisations. AI fairness assessment requires consideration of multiple factors

✉ Avinash Agarwal
avinash.70@gov.in

Harsh Agarwal
harshaga@adobe.com

1 Telecommunication Engineering Centre, Ministry of Communications, Government of India, New Delhi, India

2 Adobe Inc., Noida, India

at different stages of the AI development lifecycle [12]. There is an urgent need to create a standardised holistic framework that addresses the bias concerns of various AI systems. The framework should include independent fairness ratings, transparency reports, and disclosures.

However, standardising the process of assessing the fairness of AI systems is a challenging task. There is a lack of consensus on the definition of fairness in AI. For instance, Refs. [11, 13] each explore more than twenty definitions of AI fairness. Also, there is no widely accepted framework for assessing bias in AI systems. AI fairness is context-sensitive [14–16]. Developers use different types of machine learning (ML) techniques and algorithms for different types of AI systems. Even two similar types of systems require different approaches as each has a unique problem statement and a unique set of requirements. The consequences of biases vary for different scenarios. The sources and type of biases are also not the same for all applications. The tolerance for biases varies from case to case and also from region to region. The concept of fairness is region and culture-dependent. Something considered fair in one country or culture may be regarded as biased in another. For instance, [17] presents a study involving children from three different cultures (Kenya, Namibia, and Germany) with diverse fairness norms to demonstrate that ideas of distributive justice based on merit are not culturally universal. Another study by Charisi et al. [18] presents the difference in the perception of children in Japan and Uganda on the notion of fairness in everyday, imaginary, and robot-related scenarios. Further, fairness is also data-dependent. A face recognition model trained with European faces is more likely to be unbiased when deployed in France than in India. Terhorst et al. [19] demonstrate that popular face recognition models have biases related to non-demographic attributes also, such as accessories, hairstyles, face shapes, or facial anomalies. This makes it difficult for governments, sector regulators, deployers, and users to determine how fair an AI application is.

Considering the varied requirements for each AI application, a single monolithic end-to-end standardised process for assessing fairness may not be feasible. However, most AI systems have similar development lifecycles, from framing the problem statement to their deployment and use. Standardisation of bias mitigation procedures, fairness metrics, etc. is more pragmatic at each lifecycle stage, vaguely similar to the approach followed by the International Organization for Standardisation (ISO) seven-layer Open System Interconnection (OSI) model for data communication [20]. The OSI model divides the communications between a computing system into seven distinct layers, each layer performing a specific function. This division into layers allows for a clear separation of concerns and makes it easier to understand, design, and troubleshoot communication systems. When combined, each layer contributes to enabling full data communication.

To address the challenges of assessing the fairness of AI systems, we propose a seven-layer model for standardising AI fairness assessment. This model provides a holistic view of assessing bias and gives due importance to all stages of the AI development lifecycle. The seven layers of the model are (1) requirements, context, and purpose, (2) data collection and selection, (3) data pre-processing and feature engineering, (4) algorithm building, (5) AI system training, (6) independent audit and fairness certification, and (7) usage. Each layer represents a critical stage in the AI development lifecycle and addresses a specific set of fairness considerations. For each layer, we propose a checklist of fairness assessment questions to identify and address potential sources of bias.

While the developers play an important role in bringing fairness to an AI system, other stakeholders are equally concerned about the model fairness [21, 22]. These may include the top management of the developer organisation, the deployers, the users, governments, sector regulators, and other sections of society. Studies such as [23] show the impact on AI fairness due to a lack of engagement of the AI practitioners with the direct stakeholders or domain experts. Our proposed model aims to standardise the assessment process and enable independent audits for the benefit of all stakeholders.

For each lifecycle stage, we elaborate on the possibilities of biases, the precautions to avoid these biases, and the checklists to formalise the process. Additionally, the proposed layers can be evaluated for bias independently, allowing flexibility. The reference to the OSI model is for a better appreciation of the concept of discrete layers only. While all seven layers of the OSI model are involved simultaneously when a computer network is in operation, the layers in AI systems are related to the model-building stages.

The proposed model is motivated by the recognition that AI fairness is a complex problem that requires a multi-disciplinary approach [24]. The proposed model is also motivated by the need to promote transparency and accountability in AI systems [25, 26]. In this paper, we explain the rationale behind the proposed model, discuss its key components, and provide examples of its practical implementation. We further validate our model by examining a case study.

1.1 Gap analysis and our contribution

Most researchers focus on a few aspects of fairness. We observe a gap in a holistic evaluation of fairness covering all activities from concept to usage of the AI system. Excessive focus on biases in the datasets, fairness metrics, and algorithmic bias may lead to ignoring other aspects affecting AI fairness, such as context, deployment environment, etc. Our work attempts to fill this gap. To the best of our knowledge, we are the first to propose a seven-layer model to ensure fairness at every stage of the AI application lifecycle. Despite the differences in the various aspects, most AI systems have similar model-building stages. The proposed model splits the AI system lifecycle into seven abstraction layers, each corresponding to a well-defined AI model-building or usage stage. We also provide checklists for each layer and deliberate on potential sources of bias in each layer and their mitigation methodologies. This work will facilitate layer-wise standardisation of AI fairness rules and benchmarking parameters.

Our proposed seven-layer model is unique in several ways. First, it provides a clear structure for assessing fairness at different stages of the AI lifecycle, which enables the identification of sources of bias at each stage, allowing for a more comprehensive and structured assessment of bias. Second, the checklists proposed for each layer of the model are based on the latest research in AI fairness and align with existing workflows. Third, our model provides a clear framework for organisations to formalise the ad-hoc processes followed by developers and make them more consistent across different projects. Fourth, it enables layer-wise standardisation of AI fairness rules and benchmarking parameters.

1.2 Impact statement

Our model can be used across different domains and organisations. It covers all stages of the AI development lifecycle and addresses a wide range of fairness considerations. The checklists provided at each layer of the model will make it more efficient for the auditors to check for fairness in the AI system. By providing a clear and standardised process for assessing bias, the proposed model will help to build trust in the technology and ensure that it is deployed responsibly and ethically. This paper aims to empower designers, developers, evaluators, and service providers of AI applications to detect bias efficiently and develop fair AI systems. This paper not only helps standardise the fairness process but also makes it more thorough. It will also help individual developers, startups, and small organisations develop high-quality and fair AI systems competing with big organisations. Socially, it helps build unbiased public service applications. Legally, it helps develop a policy framework as the governments and regulators can specify layer-wise AI benchmarks. Institutions procuring AI applications can ask for layer-wise benchmarks and ratings as eligibility criteria.

2 Literature review

AI fairness is a rapidly growing research topic. In this section, we review the contemporary work related to biases observed in AI systems used by governments and leading corporations, biases due to imperfect datasets, biases due to algorithms, the concept of fairness, and metrics used to assess fairness. We also review works focusing on the typical AI lifecycle and standardising the bias-mitigating processes.

AI bias Data is the direct representation of society and the system that generates it. It is a well-known fact that society has not been fair to everyone historically. Even in today's times, discrimination is prevalent [27–29]. When one trains AI systems using datasets representing such a society without proper checks, the systems are also highly likely to be biased. Instances of various deployed AI systems having biases based on demographics, such as race, biological sex, age, etc., are well documented. Examples of racial bias include the AI system used by the US judiciary to predict the likelihood of a criminal committing a crime again [30] and AI systems used by several lenders in the US granting loans to non-eligible white Americans while marking many eligible African Americans ineligible [31]. Manrai et al. [32] demonstrates how the exclusion of African Americans resulted in their misclassification in clinical studies. Geolocation bias was observed in two of the most commonly used image datasets, ImageNet and Open Images, as the datasets were Amerocentric and Eurocentric [33]. Kodiyan [34] and others cite gender bias in the AI system used by Amazon to filter job applications as a result of training the AI system using data of historical hiring at Amazon when most of its employees were male. AI systems also exhibit age bias as training datasets often under-represent older people. In 2017, many hiring platforms were under investigation as they effectively excluded older applicants by not allowing them to enter the year of graduation before 1980 [35].

Datasets are not the only source of bias, as other activities are also prone to biases. AI system's algorithm design, training, and validation phases can also contribute to bias in the overall system [36]. Algorithm bias in customer management AI systems can be reduced using two approaches: priori and post hoc approaches [37].

Different biases observed in machine learning systems include measurement bias, aggregation bias, sampling bias, emergent bias, social bias, and algorithm bias [38]. There could be three causes of biases: bias in modelling, bias in training, and bias in usage [6]. Based on the source, biases can be classified into three categories: preexisting, technical, and emergent [39]. Biases can also be categorised into systemic, statistical, and human biases [40].

We identify different sources and types of biases that might enter the AI system at different life stages of the AI system. We further attempt to provide some measures that help mitigate biases at each layer.

AI fairness definitions and metrics Fairness metrics play a crucial role in assessing bias in AI systems and mathematically quantifying the level of fairness in an AI system by comparing the outcomes for different groups of individuals. Castelnovo et al. [41] provides a detailed comparison of different fairness metrics and definitions. Commonly used fairness metrics include demographic parity, equal-opportunity [42], equal–mis-opportunity [43], average odds, and distance metrics [44]. Demographic parity measures whether the probabilities of favourable outcomes for privileged and unprivileged groups of data are the same or not. While equal-opportunity makes sure that the model has the same true positive rates for both groups, equal–mis-opportunity makes sure that the false positive rates are the same. Average odds is the combination of equal-opportunity and equal–mis-opportunity metrics. Our model calls for fairness metrics in various layers, especially layer 5.

As each metric measures a different aspect of the data and the results, the choice of fairness metric used should be informed by the specific context and goals of the AI system and not just by the metrics themselves. The choice of the metric also depends on the fairness definition applicable. Different researchers define fairness in AI differently depending on the context of the problem statement. Verma and Rubin [11] and Narayanan [13] present a compilation of various definitions of fairness. The fairness definitions broadly fall into three categories: individual fairness, group fairness, and subgroup fairness [38].

Combined rating metrics (e.g. Fairness Score and Bias Index [45]) combine the results of various fairness metrics into a single score to provide the overall fairness rating of the AI system. Our model uses such rating metrics in layer 6.

AI lifecycle stages Next, we review the published work on the AI lifecycle. Three standard reference methodologies for the lifecycle of Machine Learning applications are Cross-Industry Standard Process for Data Mining (CRISP-DM), Knowledge Discovery in Databases (KDD), and SEMMA. CRISP-DM [46] breaks down the model into six stages (business understanding, data understanding, data preparation, modelling, evaluation, and deployment). KDD [47] considers five stages of knowledge discovery (selection, preprocessing, transformation, data mining, and interpretation/ evaluation). SEMMA also considers five stages for the process (sample, explore, modify, model, and access). Azevedo and Santos [48] and Shafique and Qaiser [49] provide comparative studies of these three lifecycle methodologies.

Different researchers classify the various activities involved in an AI lifecycle into different numbers of stages. For instance, researchers from Microsoft studied the processes and practices followed by different internal teams while working on AI problems [50]. Accordingly, they divided the machine learning workflow from requirements to model monitoring into nine stages and presented an ML process maturity metric for self-assessment of work progress. De Silva and Alahakoon[51] presents the AI life cycle as consisting of three phases (design, develop, and deploy) and nineteen constituent stages across the three phases from conception to production. Wang et al. [52] examines the product lifecycle management in three stages (product design, manufacturing, and service) with four sub-stages for each stage and maps the AI technical requirements to each stage.

These references provide insights into the AI lifecycle stages but do not cover the fairness or bias mitigation aspects. While the different studies cover similar activities/ processes during the AI lifecycle, there is a lack of consensus on classifying the various activities/ processes involved into distinct lifecycle stages. We build on the previous work and propose to classify the AI lifecycle into seven distinct abstraction layers.

AI lifecycle - bias and auditing Previous work identifies types of potential biases associated with each lifecycle stage. For instance, [53] considers the machine learning lifecycle in seven steps and identifies seven distinct sources of bias. Fahse et al. [54] identifies eight distinct types of bias and allocates them to the six phases of the CRISP-DM model. We use these inputs as we frame checklists for each layer of our model.

Bantilan [55] proposes a Fairness Aware machine learning interface called themis-ml for discovering discrimination in ML. The proposal has four interfaces each corresponding to the four components of the AI classification pipeline. The paper focuses only on simple binary classifiers and does not cover all the layers such as data ingestion.

Published works related to AI lifecycle audit and assurance frameworks provide insights about assessment procedures. Ashmore et al. [56] presents various assurance methods while considering the machine learning lifecycle into four stages (data management, model learning, model verification, and model deployment). Raji et al. [57] proposes an end-to-end auditing framework for developing AI systems to increase accountability. It suggests generating documents at each phase of the development lifecycle that is then internally audited and scrutinised. Through a case study of AstraZeneca, [58] presents the limitations and challenges faced by organisations undergoing ethics-based AI audits. These works primarily focus on the overall audit, not specifically on AI fairness. We consider these inputs in framing the fairness assurance questionnaire for each layer of our model.

AI ethics checklists and guidelines Furthermore, studies such as [59–61] propose checklists for ensuring fairness in AI systems. While AI fairness checklists are important in organisations to formalise ad-hoc processes, the checklists need to be aligned with the existing workflows to avoid their misuse [62]. Richardson and Gilbert [63] reviews various popular software toolkits and checklists as the solutions to tackle algorithm bias. Seedat et al. [64] provides checklists at the four stages of the ML pipeline: Data, Training, Testing, and Deployment. The checklists aim to evaluate and monitor the data so that the ML model is reliable.

Some of these checklists are for specific scenarios only. Further, these checklists are generally focused on primarily assisting the developer in achieving AI fairness. Our model aims to provide checklists for all lifecycle stages and standardise the process, which would help other stakeholders also.

Guidelines on ethical or trustworthy AI published by various organisations and researchers give valuable inputs for standardising the fairness assessment process. Ryan and Stahl [65] and Jobin et al. [66] provide a comprehensive overview of various AI ethics guidelines. Hleg [67] by High-Level Expert Group on Artificial Intelligence (AI HLEG) furnishes a detailed checklist of guidelines for the developers working on AI systems to self-assess and ensure they develop along the principles of Trustworthy AI. Researchers have also proposed frameworks for trustworthy AI systems focussing on the ethics of data and ethics of algorithms [68] and common AI fairness principles for building guidelines for use by all the stakeholders involved [69]. We build upon these studies to provide a comprehensive framework covering not only the development lifecycle but also the deployment phase.

Our paper is a continuation of all these efforts by various researchers to identify biases and mitigate them to develop fair AI systems. Most fairness metrics use the predictions/ outputs of the AI model to evaluate its fairness. As a result, developers relying primarily on the fairness metrics learn about the model's fairness late in the development cycle, leading to more effort and cost. We provide a layered approach and checklists to tackle biases in the different phases of an AI system development lifecycle, thus allowing the stakeholders to take corrective measures from the beginning.

3 The seven-layer model

The proposed model consists of seven layers based on the AI lifecycle. The roles of developers and deployers/ users, and the types of potential biases at each layer are tabulated in Fig. 1 as follows:

3.1 Layer 1: Requirements, context, and purpose layer

The first layer pertains to understanding the requirements, context, purpose, and concerns of the proposed AI system. This layer does not involve data handling or coding but is still essential in building a fair and unbiased AI system.

It involves understanding why AI is required to solve the problem statement. Knowing the current solution in place helps in better understanding the problem. The developers and the managers should brainstorm on the requirements. Knowledge of the domain allows the developers to grasp the possible biases they might encounter as they start development.

The scope and context of the AI system should be clear. There should be clear documentation of the expectations from the AI model, the intended users, and the anticipated deployment environment. The developers decide the machine learning model type best suited for the instant usage case. They also identify if datasets are already available for model training.

There should be due diligence for fairness. The service providers and the developers should identify the attributes that are prone to biases. There should be deliberation on the tolerance/acceptable limit for biases in the AI system. The demography of the users will lay the foundation for expected biases in the data and help the developers to take precautionary measures in the early stage of the development. Reviewing relevant AI literature helps in identifying any previously recorded instances of biases while solving similar problems.

This layer also involves identifying the AI fairness technique, such as group fairness or individual fairness, which will be used throughout the rest of the layers to identify biases in the AI model. There are multiple definitions of AI fairness, each having its own set of fairness measuring metrics. Identifying the technique at an early stage sets an effective measure to detect biases quickly.

Fairness checklist for this layer

1. What are the expectations from the AI model?
2. Are the requirements clear and formally documented?
3. Have the developers understood the scope and context of the problem?
4. How is the problem being dealt with presently?
5. Which fairness technique is appropriate and why?
6. How are the tolerance limits for fairness and bias decided?
7. Was the relevant AI literature reviewed?
8. Is the data generated internally or from external third-party/ public sources?
9. Were benchmarks and baselines identified?
10. Are Pre-trained models already available that can be repurposed?
11. What are the alternate solutions and flows?
12. Are the stakeholders, domain experts, and auditors identified?
13. Does the stakeholder group have a proportionate representation of all demographic groups that might be affected by this model?

Layer	Layer Name	Roles of Developers	Roles of Deployer/ Users	Types of Biases
1	Requirements, Context, and Purpose Layer	Understand why AI is required; identify potential biases, data sources, users, and demographies; cost v/s tolerance analysis	Explain context, use case, anticipated biases; suggest data sources, protected attributes. Explain system requirements (tolerance, privacy, explainability, etc.)	Framing effect bias, stakeholder bias
2	Data Collection and Selection Layer	Identify data sources - availability, representativeness, suitability; data labelling		Historical bias, population bias, selection bias, social bias, behavioural bias, temporal bias, confirmation bias
3	Data Pre-processing and Feature Engineering Layer	Decide protected attributes; pre-process training data; take care of risks of oversampling, outliers, synthetic data, privacy (anonymization/ de-identification)	Decide fairness required for group or individual or both; decide protected attributes	Measurement bias, omitted variable bias, representation bias, aggregation bias, sampling bias, linking bias
4	Algorithm Layer	Decide ML technique and algorithm; deploy diverse teams to reduce biases of developers		Developer bias, algorithm bias, popularity bias
5	AI System Training Layer	Model training; check fairness, explainability, accuracy, etc.; check bias for each protected attribute using fairness metrics; Overall fine-tuning		Evaluation bias, behavioural bias, presentation bias, content production bias, social bias
6	Independent Audit Layer	Benchmarking, third-party/ self audit following standard processes. Rating metrics (e.g. Fairness Score)		Evaluation bias
7	Usage Layer	Periodically check for usage bias; periodic retraining	Ensure usage as per context, no skewed use. Provide feedback on any bias; incident reporting	Deployment bias, usage bias, emergent bias, user interaction bias

Fig. 1 The seven-layer model for standardising AI fairness assessment and the types of biases [38–40]

14. Were fairness-related potential harms identified and discussed?
15. Who would be the beneficiaries of this model and will it negatively affect any group/ individual?
16. Are there any relevant standards, policies, or guidelines?
17. Have all the project members had ethical training or attended workshops/courses concerning AI ethics?

3.2 Layer 2: Data collection and selection layer

The data collection and selection layer comprise activities related to gathering data required for training the AI system. This layer includes identifying potential data sources, creating the data collection pipeline, and labelling the data based on the problem statement.

If the input data has biases, such as social biases, then the AI system trained on it will reflect the same biases. Thus, detecting and removing biases right at the initial stage are essential to ensure that the dataset is fair. Analysis of several AI-related biased outcomes establishes that biased training data is the foremost cause of bias in AI systems as the algorithms learn from the patterns in the training dataset [70].

The data collection process may introduce bias, even though the dataset is not biased. Often, data is either not labelled or needs to be labelled differently based on the context of the problem statement. The process of labelling data can be time-consuming and labour-intensive, and companies frequently outsource the labelling job to specialised companies or individuals called data annotators, such as Amazon Mechanical Turk [71]. Labelling can also be done by in-house teams or volunteers. The labelling process is prone to introducing bias in the system, as the understanding and personal perceptions of the individuals labelling the data decide the labels assigned to the data. A recent MIT study

[72] found thousands of mislabeled samples in the datasets used to train commercial systems.

It could be possible that the data collected is not balanced because of selective filtering/sampling of data. Sampling bias can cause poor generalisation of the learning algorithms [73]. Developers might leave out important data because of either their prejudice or ignorance of its importance.

A critical issue is unbalanced data at the source. People belonging to a certain race, biological sex, or community are often under-represented, as compared to others, in various datasets because of historical discrimination or restricted access to services. This disparity in data quality is also a cause of algorithm bias [74]. Also, data of certain groups would be incomplete or inaccurate compared to other groups, leading to more noise in the dataset for such groups.

Too much data is also harmful. AI systems can make better predictions with less information too. The fundamental point is that whatever information is available is relevant to the problem statement. To increase the training dataset size, developers sometimes add extra information that might not apply to the instant case, leading to information bias. Also, to expand the dataset size, data might be collected from unverified sources, thus raising the risk of bias in the resultant AI system.

Data captured using defective or faulty devices can lead to measurement/device bias [75], where the data is systematically inaccurate or inconsistent. Such data causes the model to learn patterns that do not exist in the true underlying data and lead to poor performance on new, unseen data. Instead, it may be necessary to remove or correct the bias in the data, before training the model, by identifying the defective devices and removing the data they generated, or by applying correction factors to the data to compensate for the bias. For example, a poorly calibrated Internet of Things (IoT) sensor can lead to false alerts, and a system trained using its data will be biased and inaccurate.

Data collected from unverified sources can lead to bias in AI models. For model training, datasets that can be or have been audited independently by domain experts should be preferred. While using public datasets, following the data collection and annotation protocols by the organisation/individual collecting the data should be ensured.

Thus, systematic selection of appropriate variables and collection of relevant data is necessary to reduce algorithm bias [63].

We suggest various checks to ensure that the data collected and selected for system building is fair and unbiased.

Fairness checks while selecting a dataset

1. Is the dataset collected from a known and verifiable data repository?
2. Is there a way to audit the dataset for validity and accuracy?
3. Is the dataset complete? Are there too many missing values?
4. Is the dataset cleaned and pre-processed or raw?
5. How distributed is the dataset? Does it include data from all sections of society? Is it skewed in favour of certain groups?
6. Is there any metadata available along with the dataset that increases its explainability? Is the information available about who collected it, when it was collected, and the collection processes used?
7. Why was this data collected?
8. How old is the dataset? Do the situations/assumptions at the time of data collection still apply?
9. Is more than one dataset being considered? If yes, what is the correlation between them?
10. Are all the attributes in the dataset legally approved for use?

Fairness checks while collecting and labelling data

1. Is the dataset already labelled?
2. In the case of manual labelling, are the people labelling the data trained and well aware of the context of the problem?
3. Are quality assurance and verification processes included to ensure no developer bias gets into the dataset?

3.3 Layer 3: Data pre-processing and feature engineering layer

Data pre-processing and feature engineering form the next layer. Most of the datasets used for training the model need pre-processing. The same dataset can train multiple AI systems, each having a different problem statement, requiring different input data formats, and/or requiring separate labels. Some AI systems might require aggregated data, while others might work on a different numeric scale altogether. Some AI systems might need only a subset of records or use only a few features.

Pre-processing is the set of steps required to make the raw data suitable for analysis and training. It primarily involves assessing data quality, cleaning the data, and data transformation. Depending on the problem statement and the dataset quality, one might have to remove outliers, correct data type mismatches, remove null values or fill them with synthetic values, and perform normalisation, among other pre-processing steps.

Bias can easily creep into the system in this layer. Aggregating and normalising the records can lead to measurement and aggregation biases. Important features might get sidelined or scaled down to a level that they lose significance, leading to omitted variable bias [76] and exclusion bias.

Too much synthetic data is not a good representation of the ground reality and can lead to unfair predictions by the AI system. Synthetic data can lead to overfitting. Overfitting occurs when a model is trained to fit the training data too closely, resulting in a model that performs well on the training data but poorly on new, unseen data. When synthetic data is used to train a model, the model may learn features that are specific to the synthetic data and not generalise well to real-world data. Overfitting can lead to bias and compromise fairness if the training data is not representative of the real-world population. For example, as per the American Time Usage Study (ATUS) [77], the average sleep duration of an individual is linked to their age. Synthetic data based on this dataset must correctly capture the proportion of subjects in each age group and accurately represent the sleep habits within each subgroup, such as race, gender, school activity, etc. /citepbhanot2021problem.

Removing rows with null values or outliers can lead to a significant drop in the number of records of a particular demography/section of society that, because of historical reasons and discrimination, might not have had that much representation in the dataset, leading to fewer and improper records.

The identification of protected attributes and privileged classes for each protected attribute is the first step toward developing a fair AI system. Protected attributes are usually those features that are associated with social bias. Some common protected attributes are biological sex, race, caste, religion, etc. Protected attributes are application dependent; a protected attribute for one application may not be so for another. For example, biological sex may be a protected attribute for a recruitment application but not for a medical application, such as breast cancer patient forecasting. Protected attributes should be decided meticulously not just by the developers, but also involving other stakeholders, legal experts, data scientists, and sector regulators, wherever required, to ensure that the developer's personal bias does not play into the picture.

Another component of pre-processing is bias mitigation. If there are already known biases in the dataset, we use mitigation techniques to reduce their influence on model training to a certain extent. Techniques, such as sampling, reweighing, and data transformation, are commonly used to reduce the effect of biases in the dataset [78, 79]. Sampling techniques such as stratified sampling ensure that the dataset selected for training has the same proportion of each subgroup as represented in the population. Reweighing is used to adjust the importance of different examples in the dataset so that under-represented groups are given more importance. Data transformation techniques, such as adversarial debiasing, remove or reduce the influence of sensitive attributes in the dataset. Overall, these techniques can help reduce the impact of bias in the dataset by making it more representative of the population and reducing the influence of sensitive attributes on the model's predictions.

Fairness checklist for this layer

1. What are the approaches taken to clean the dataset? Are they leading to an imbalanced dataset?
2. How is feature engineering done? Was it done by a single developer or reviewed by others also?
3. Were any features dropped? If yes, why and what effect do they have on the problem?
4. Would the pre-processing process remove an enormous chunk of data from a particular demography?
5. As part of over-sampling, did the algorithm add too much synthetic data?
6. How were the normalisation and scaling techniques decided?
7. How were the protected attributes and privileged classes selected?

3.4 Layer 4: Algorithm layer

The Algorithm layer is a set of activities related to writing the code for the model logic. It involves selecting the most appropriate machine learning type, algorithm, and fairness criteria and writing the code.

This layer is susceptible to developer bias at several junctures. All developers perceive fairness in their way. If not checked, their individual beliefs will affect the overall output of the models they build. As a solution, companies are increasingly using automated tools to make accurate and comprehensive choices. However, as [80] noted, while reliance on technology may avoid human bias, the potential to produce biased algorithms opens up a dark side of algorithm-based decisions.

The machine learning technique used, out of supervised, unsupervised, semi-supervised, or reinforcement learning, has an enormous impact on the overall fairness of the AI system. For a given case, all the ML techniques are not equally appropriate. Some provide good results in one situation but lead to overfitting in another. Overfitting around the privileged class leads to an unfair AI system that gives biased results against the unprivileged class. The Robodebt scheme in Australia wrongly and unlawfully pursued hundreds of thousands of welfare clients for the debt they did not owe [37, 81]. Biased vehicular automation may cause severe damage, as evident in the accidents caused by the self-driving cars by UBER [82] or in the incident of deaths caused by the malfunctioned robot at an Amazon warehouse [83].

Many algorithms involve auto-encoders that reduce features of the dataset so that it becomes easier to learn. This feature extraction process can introduce bias as it might discard contextual information that could be relevant in some situations [84].

The black-box nature of AI algorithms makes it difficult to debug the root cause behind bias in the AI system. Developers should attempt to make the AI system development process as transparent and explainable as possible. Model interpretability should be a factor when selecting an algorithm. One should always prefer simple models over highly complex black-box models. Angelino et al. [85] and other studies observe that the performance of several simple models is as good and accurate as the state-of-the-art models.

This layer also includes the identification of the preferred choice of fairness technique. There are multiple ways to define and analyze the AI system fairness, such as individual fairness and group fairness [86].

Fairness checks during the development phase

1. Is the development process transparent?
2. Are there regular reviews and quality assessments in the organisation to check that the developers' individual biases do not influence their work?
3. What were the criteria for selecting the algorithm technique?
4. Were the stakeholders consulted? Was the context of the problem statement taken into consideration?
5. Which fairness technique was selected—individual fairness, group fairness, or something else?
6. Is the model developed per the requirements specified in layer 1?

3.5 Layer 5: AI system training layer

The AI system training layer relates to activities associated with the ML model training. This stage includes dividing the dataset into training, testing, and validation sets, training the model iteratively, and evaluating its fairness using fairness metrics.

Model training is an iterative process, where after each iteration, the developers evaluate its performance using the testing dataset and determine how many further iterations are required. Finally, they use the validation dataset to determine the expected performance of the model in the field. Thus, distributing data in the three datasets without prejudice is crucial for the quality of the model. Even if the input dataset is unbiased, a biased or inappropriate splitting of it into the three datasets could lead to biases in the AI system. While a biased training dataset leads to a bias in the trained model, a biased testing or validation dataset does not fairly evaluate the performance of the trained model.

One of the steps in model training is hyperparameter tuning. The goal of the hyperparameter tuning process is to find the best set of hyperparameters that results in the highest accuracy on the training data. If the tuning process is not carried out systematically in a controlled manner, the model might overfit the training data and not generalise on unseen data. If the developers are not careful, test data may also get considered while deciding the hyperparameters, leading to an inaccurate model. Further, different developers interpret the results differently. For example, different developers might select different learning rates based on their interpretations. Another example is that they might have a personal bias in favour of one optimization function over another. All these decisions decide how the hyperparameters are tuned.

For the same problem statement, different developers might consider separate parameters to represent the model performance. It could be accuracy, precision, recall, f1 score, or something else. There could be scenarios where recall is more important than precision or vice versa. For example, if the aim is to select and publish the top ten positive user reviews for a hotel, the model should detect all positive reviews. It may mark some negative reviews as positive with a low score (i.e. higher recall) instead of leaving out some positive reviews to avoid some negative ones getting marked as positive (i.e. higher precision). Thus, correct performance parameter selection is essential and directly impacts the model's fairness.

An overfit or an underfit model will likely be unfair and biased towards one data section. The trade-off between fairness and performance also reflects individual and organisation bias.

Selecting inappropriate benchmarks for comparing the model performance may lead to evaluation bias. For example, some benchmarks used to test facial recognition systems, such as Adience and IJB-A, are known to be biased toward skin colour and gender [73].

After training the model, check fairness for each protected attribute. One should evaluate the fairness of both the training dataset as well as the model outcomes for the test dataset. Selecting the most appropriate fairness metrics for each protected attribute is crucial. The developers may use multiple fairness metrics for each attribute. Some commonly used fairness metrics are statistical parity difference, disparate impact, equal opportunity, equal–mis-opportunity, and average odds.

Fairness checks during the training phase

1. How was the dataset divided into training, testing, and validation sets? Was the data distributed to them randomly?
2. Were the resultant three datasets balanced? Did they have an equal/proportional distribution of privileged and unprivileged classes?
3. How were the hyper-parameters selected?
4. What are the fairness metrics used, and why?
5. Is fairness checked for each protected attribute or not?
6. What are the identified benchmarks, parameters, and metrics for measuring model performance, and why?

7. The choice of metrics, parameters, and benchmarks should be well documented and available for external auditing and explainability.

3.6 Layer 6: Independent audit layer

The Independent Audit layer includes testing and fairness certification by a team other than the developers. It requires a standardised fairness assessment process, a rating metric independent of the various fairness metrics used in layer 5, and independent auditors.

With the increasing use of AI in all aspects of our lives, the users want to know that the decisions that the AI systems make are fair or not [45]. Demand for independent audits based on a standardised process and certification or rating of AI systems before deploying them is fast increasing. A standard certification process is required to increase trust among the users of the AI model.

Various governments are using AI applications to deliver public services and identify beneficiaries of social welfare schemes. Such applications must not be biased against or in favour of some groups of citizens. The judiciary is fast adopting AI systems to assist in court cases. Investigation agencies use AI to monitor online crime, facial recognition, and threat prediction, among many other use cases. Smart cities, traffic control, banking, and many other public services are incorporating AI systems at various levels to provide better facilities. With the increasing convergence of AI and public services, AI fairness certification would soon be a legal requirement, not just an ethical concern.

Researchers have proposed auditing to verify the fairness, bias, and validity of AI-driven predictions [57, 87]. A well-conducted audit can identify adverse effects, suggest corrective measures, and help decision-makers maximise the value of the model. Auditing should occur throughout all stages of model development [88]. Documentation of these efforts can minimise the need for post hoc auditing. Good quality auditing can increase public trust in the AI model, the deployer, and AI technology in general.

Independent audits and certification also help in increasing the model's explainability and provide transparency. Additionally, the service providers get another factor to consider while comparing and selecting AI models for their services. Governments and multilateral agencies following a transparent procurement process could use the fairness certificate as an eligibility criterion.

Agarwal et al. [45] propose a standardised process for assessing the fairness of supervised-learning AI systems and rating them on a linear scale. Accredited third-party agencies or a separate in-house team not involved in the model building may do the audit following the standardised process.

The auditor should evaluate fairness for each protected attribute. Fairness metrics used may vary as per attributes and also usage cases. Uniformity and comparison between different AI models require a universal standard rating metric. This metric should derive from the individual fairness metrics used in layer 5, but its score should not depend on any specific underlying metric used in layer 5. In other words, the score of the combined rating metric should not change much if a different set of fairness metrics is used at layer 5. It should also give a single score for the overall model instead of separate scores for each protected attribute. The Fairness Score and Bias Index proposed by Agarwal et al. [45] are examples of such combined metrics suitable for measuring the overall fairness of the AI system. Bias Index gives a single score indicating the bias for each protected attribute by combining various individual fairness metrics used for that protected attribute and is not dependent on the number or type of such metrics used. Fairness Score gives a single score for the entire model by combining the individual fairness metrics used for all the protected attributes in the dataset, again without depending on the number or type of such metrics used.

Fairness checklist for this layer

1. Is the auditor independent? If it is an in-house team, then is it different from the developers? If it is a third-party auditor, then is it accredited?
2. Is a standard fairness assessment process followed?
3. Is the rating matric universal?
4. Has the auditor checked all the protected attributes?

3.7 Layer 7: Usage Layer

The final layer is the Usage Layer. Here, the AI system moves from development and testing to commercial use. This layer involves deploying the AI system in the field, its regular monitoring and periodic retraining, and timely maintenance.

Deploying an AI system designed and trained for a particular situation in a different setting could adversely affect not only its business but also its fairness. AI systems trained for facial recognition with the dataset of Asian faces might not be accurate and fair if deployed in Europe or the USA.

Data is dynamic; the ground reality keeps on changing. A dataset used in training the system may, after some time, no longer represent the prevailing situations. It is also possible that the business goals shift or there is a change in the underlying technology. All these factors might cause an AI model, fair at the time of deployment, to give biased results.

There is a need for regularly retraining the model while accounting for the changing situations. The retraining keeps the model updated with the latest trends, leading to better accuracy and fairness. There are two ways to use the new data gathered since the last system training for the AI system retraining. One way is to add the new data to the old data and retrain the system on the complete data. The other way is to discard the old data and use only the newly gathered data for system retraining. One can select either of the two approaches based on the usage case, data generation speed, and the time gap between the two training sessions.

Before retaining, we should perform fairness checks on the new data as a skewed use of the AI system may lead to biases in such new data.

After deploying the system, regularly monitor its performance using the fairness metrics used in the previous layers as Key Performance Indicators (KPIs). Any drop in the metrics should be an early sign of the need for system retraining.

Fairness checklist for this layer

1. Are fairness KPIs regularly monitored after the release?
2. How much time has passed since the last round of model training?
3. Have the underlying ground realities changed since deploying the model?
4. Is the distribution of protected classes in the protected attributes the same in production data as in the training data?
5. What were the deciding factors for retaining or discarding the old data while retraining the model?
6. Are there mechanisms in place to address data shifts?

4 Bias mitigation strategies at each layer

The broad bias mitigation strategies at each of the seven layers are summarised as follows:

Layer 1: Bias mitigation starts at the first stage itself with due diligence about AI fairness. The developers should clearly understand the scope of the problem statement, the context, and the expectations from the AI system before the actual development starts. Identifying data sources, protected attributes, possible privileged and unprivileged classes, ML techniques, fairness models, and understanding tolerance limits to bias helps build a fairer AI system.

Layer 2: The most common source of bias entering the AI system is the datasets used in the model training. Biases can be avoided by selecting datasets only after verifying the data source and auditing the datasets for fairness. Selecting updated datasets will ensure that the AI model trains with the latest trends and does not learn the defunct prejudices that have been prominent in society.

Layer 3: Effective pre-processing and feature engineering not only improve the model accuracy but also reduce biases in the datasets. Bias mitigation techniques such as oversampling and adversarial debiasing have proved successful. Formally define the protected attributes and privileged classes.

Layer 4: Developer bias at the algorithm layer may be mitigated by engaging a diverse team of developers, not relying on an individual, and getting the code reviewed by another team. Confirm that the selected ML technique and algorithms are appropriate for the given case. Use simple and explainable models.

Layer 5: Splitting the input dataset into training, testing, and validation datasets in an unbiased manner, so that each dataset properly represents the target audience is necessary to ensure fair model building. Selecting the appropriate fairness metrics while iteratively training and testing the model is crucial to ascertain that the model fairness is within the tolerance limits as decided in layer 1.

Layer 6: Demand for independent audits will rapidly grow as governments and the judiciary increasingly use AI systems for citizen-centric services. A standardised fairness assessment process and a universal rating metric, as proposed by Agarwal et al. [45], are pivotal for this layer. Self-disclosures of the fairness ratings and ecosystem of accredited certifying agencies will propel awareness about AI fairness and result in better compliance with fairness requirements.

Layer 7: Biases during use can be mitigated by ensuring usage as per design assumptions, regularly checking the fairness KPIs, retraining the system whenever required, and mitigating bias from the newly gathered data before its use for retraining.

5 Case study

As a case study, we pass one of the AI systems analysed by Agarwal et al. [45] through our seven-layer model. We select the AI system using the German Credit Dataset [89] to predict the risk of loan repayment by a customer. The dataset has 20 attributes and 1000 records of German borrowers collected in the early 1990 s. The researchers identified bias in the dataset in favour of the male borrowers and tried to mitigate it using oversampling and undersampling bias mitigation techniques.

In our review, we analyse how fair it would be to deploy the same AI system to predict the loan repayment probability of borrowers in a developing Asian country. We prepare a checklist for each phase of the AI lifecycle as if the system is being developed from scratch again. We then evaluate whether the system passes the various checks elaborated for each of the seven layers of our model. The observations are tabulated in Table 1:

If we rely only on the fairness metrics (layer 5) and the overall Fairness Score (layer 6), we get a misleading impression that the application is fair. However, the checks proposed in the preceding sections for layers 2 and 7 show that the application may not be fair for the target users. The checklist indicates usage other than the intended one.

Table 1 Case study for checklist of the seven-layer model

Layer	Checklist questions	Observations
Layer 1	What are the expectations from the AI model?	The model aims to predict if a customer is a safe borrower or not i.e. can they pay back the loan?
	Is the data generated internally or from external third-party/public sources?	Data is from an external public source
Layer 2	How old is the dataset?	The AI model is trained on the German Credit Data collected in 1994. The dataset is more than 25 years old
	Do the situations/assumptions at the time of data collection still apply?	The model is trained on an old dataset. The economic conditions of the people have evolved in the last 25–30 years. Newer financial instruments and rules have been established. Hence, the dataset might not be a true representative of the current scenario
	How distributed is the dataset? Does it include data from all sections of society? Is it skewed in favour of certain groups?	The dataset is not uniform and skewed. 69% of the records are of male customers
	Is the dataset already labelled?	Yes, the dataset is already labelled
Layer 3	What are the approaches taken to clean the dataset? Are they leading to an imbalanced dataset?	Oversampling (SMOTENC) and random undersampling bias mitigation techniques were used to correct the imbalance in the dataset
	How were the protected attributes and privileged classes selected?	The attribute 'Sex' was selected as the protected attribute and the class 'male' was selected as the privileged class
Layer 4	Which fairness technique was selected—individual fairness or group fairness, or something else?	Group fairness techniques were used
Layer 5	How was the dataset divided into training, testing, and validation sets? Was the data distributed to them randomly?	The dataset set was randomly divided with 85% as the training set and 15% as the test set
	What are the fairness metrics used, and why?	Fairness metrics used: For the training dataset: statistical parity difference (SPD) and disparate impact (DI). For the model and test data outcomes: equal opportunity difference (EOD), equal–mis-opportunity difference (EMOD), and average odds difference (AOD)
	Is fairness checked for each protected attribute or not?	Yes, fairness was checked for each protected attribute
Layer 6	Did the model go through any independent or third-party audit?	No, there was no third-party audit of the AI model
	Were any rating metrics used?	Yes, the authors used the Bias Index and Fairness Score proposed by them in the paper. A fair model should have a low Bias Index and a high Fairness Score. Initially, the Bias Index and Fairness Score were 0.1037 and 0.8963, which after mitigating bias in the protected attribute, changed to 0.0864 and 0.9136, respectively
Layer 7	Have the underlying ground realities changed since deploying the model?	The context is different, and the social and economic conditions of the target audience are different from those available in the training dataset. The model is trained on the data of German loan borrowers while it is being deployed for a developing Asian country. The lifestyle of the citizens, financial institutions, regulations, economic stability, and many other factors would be very different for the two sets of borrowers

With readymade applications increasingly available, there are growing chances of using the application for demography other than for which it was developed. The case study confirms that the seven-layered model and checklists suggested for each level will ensure that the application is fair for the intended usage case and that the developers identify the requirements and challenges at the initial stages itself.

6 Conclusion and future work

AI systems are susceptible to biases at every stage, from conceiving the project to its use. This paper proposes to standardise the bias detection and mitigation process throughout the AI system lifecycle through a novel seven-layered model. Conventional approaches for tackling biases usually focus more on a few specific aspects of the AI systems, with the possibility of other elements getting ignored. A layered approach, such as the one proposed in this paper, enables a holistic view of bias handling.

The seven-layer model proposed in this paper provides a more structured, systematic, and standard approach to dealing with AI biases, required as we move from the present-day soft-touch self-regulatory conditions to a more regulated environment for AI fairness. It aims at process standardisation to enable independent fairness assessment at each layer through a standard process. The checklists proposed for each layer formalise the ad-hoc processes presently followed to check for fairness.

With the growing use of AI applications in all domains and increasing awareness of the need for fairness, customers will demand more disclosures on the fairness benchmarks. Also, governments and regulatory bodies are actively pursuing policy formulations and regulations to ensure AI fairness. This work will help individual developers, startups, and small organisations that lack institutional procedures develop fair AI systems. It will also help governments and sectoral regulatory bodies to notify layer-wise fairness benchmarks for various critical applications.

The paper proposes to split the AI system lifecycle into seven distinct layers—requirements, context and purpose, data collection and selection, data pre-processing and feature engineering, algorithm building, AI system training, independent audit and fairness certification, and usage. Layers separate from each other based on the activities involved. While the layers are interdependent, one can view each layer independently for bias handling. An analogy with the OSI framework helps to understand the concept of layers in the context of fairness. The paper also elaborates on the roles in each layer of various stakeholders, such as developers, service providers, and users.

The paper also discusses the various types of biases possible at each layer and the questions to be asked during model building to avoid these biases. Each layer uses different metrics and benchmarks to assess and measure fairness. Layer 5 (AI system training) uses fairness metrics such as disparate impact, equal opportunity difference, statistical parity difference, etc. Layer 6 (Independent audit and fairness certification) uses rating benchmarks such as Fairness Score and Bias Index.

AI fairness is context-sensitive and case-specific. A single framework for assessing fairness for all AI systems is not pragmatic. Still, various types of AI systems have similar stages in model building. The layered approach provides abstraction at each functional layer, thus facilitating the analysis and comparison of fairness requirements for different types of AI systems at the layer level. The AI system is fair if all the layers meet the layer-specific fairness requirements.

Future work To fully implement the concept of layered handling, we suggest further studies for formulating fairness benchmarks, akin to the Fairness Score, for all layers. The ethical AI rules, assessment procedures, and fairness measurement metrics can be defined for each layer, thus facilitating standardisation. The various Standard Development Organisations, AI companies, and researchers can work on developing layer-wise guidelines and standard operating procedures for assessing fairness at each layer.

Funding The authors did not receive support from any organization for the submitted work.

Availability of data and material Not applicable.

Declaration

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

Disclamer The views or opinions expressed in this paper are solely those of the authors and do not necessarily represent those of their respective organizations.

References

1. Ntoutsi, E., Fafalios, P., Gadiraju, U., Iosifidis, V., Nejdl, W., Vidal, M.E., et al.: Bias in data-driven artificial intelligence systems—an introductory survey. Wiley Interdiscip. Rev.: Data Min. Knowl. Discov. **10**(3), e1356 (2020)
2. Flores, A.W., Bechtel, K., Lowenkamp, C.T.: False positives, false negatives, and false analyses: a rejoinder to machine bias: there's software used across the country to predict future criminals. and it's biased against blacks. Fed Prob. **80**, 38 (2016)
3. Datta, A., Tschantz, M.C., Datta, A.: Automated experiments on ad privacy settings: a tale of opacity, choice, and discrimination. arXiv preprint arXiv:1408.6491. (2014)
4. Barocas, S., Selbst, A.D.: Big data's disparate impact. California law review. pp. 671–732 (2016)
5. Koski, E., Scheufele, E.L., Karunakaram, H., Foreman, M.A., Felix, W., Dankwa-Mullan, I.: Understanding disparities in healthcare: implications for health systems and AI applications.

In: Healthcare Information Management Systems: Cases, Strategies, and Solutions. Springer; pp. 375–387 (2022)
6. Ferrer, X., van Nuenen, T., Such, J.M., Coté, M., Criado, N.: Bias and Discrimination in AI: a cross-disciplinary perspective. IEEE Technol. Soc. Mag. **40**(2), 72–80 (2021)
7. Wegner, L., Houben, Y., Ziefle, M., Calero Valdez, A.: Fairness and the need for regulation of AI in medicine, teaching, and recruiting. In: Digital Human Modeling and Applications in Health, Safety, Ergonomics and Risk Management. AI, Product and Service: 12th International Conference, DHM 2021, Held as Part of the 23rd HCI International Conference, HCII 2021, Virtual Event, July 24–29, 2021, Proceedings, Part II. Springer. pp. 277–295 (2021)
8. Binns, R., Kirkham, R.: How could equality and data protection law shape AI fairness for people with disabilities? ACM Trans. Access. Comput. (TACCESS) **14**(3), 1–32 (2021)
9. Zhou, J., Chen, F., Berry, A., Reed, M., Zhang, S., Savage, S.A., survey on ethical principles of AI and implementations. In: IEEE Symposium Series on Computational Intelligence (SSCI). IEEE **2020**, 3010–3017 (2020)
10. Giovanola, B., Tiribelli, S.: Beyond bias and discrimination: redefining the AI ethics principle of fairness in healthcare machine-learning algorithms. AI Soc. pp. 1–15 (2022)
11. Verma, S., Rubin, J.: Fairness definitions explained. In, ieee/acm international workshop on software fairness (fairware). IEEE **2018**, 1–7 (2018)
12. Toreini, E., Aitken, M., Coopamootoo, K., Elliott, K., Zelaya, C.G., Van Moorsel, A.: The relationship between trust in AI and trustworthy machine learning technologies. In: Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency. pp. 272–283 (2020)
13. Narayanan, A.: Translation tutorial: 21 fairness definitions and their politics. In: Proceeding Conference of Fairness Accountability Transport, New York, USA. vol. 1170. p. 3 (2018)
14. Abu-Elyounes, D.: Contextual fairness: a legal and policy analysis of algorithmic fairness. U Ill JL Tech & Pol'y. p. 1 (2020)
15. Mohseni, S., Zarei, N., Ragan, E.D.: A multidisciplinary survey and framework for design and evaluation of explainable AI systems. ACM Trans. Interact. Intell. Syst. (TiiS) **11**(3–4), 1–45 (2021)
16. Mulligan, D.K., Kroll, J.A., Kohli, N., Wong, R.Y.: This thing called fairness: disciplinary confusion realizing a value in technology. In: Proceedings of the ACM on Human-Computer Interaction. 3(CSCW):1–36 (2019)
17. Schäfer, M., Haun, D.B., Tomasello, M.: Fair is not fair everywhere. Psychol. Sci. **26**(8), 1252–1260 (2015)
18. Charisi, V., Imai, T., Rinta, T., Nakhayenze, J.M., Gomez, R.: Exploring the concept of fairness in everyday, imaginary and robot scenarios: a cross-cultural study with children in Japan and Uganda. In: Interaction Design and Children. pp. 532–536 (2021)
19. Terhörst, P., Kolf, J.N., Huber, M., Kirchbuchner, F., Damer, N., Moreno, A.M., et al.: A comprehensive study on face recognition biases beyond demographics. IEEE Trans. Technol. Soc. **3**(1), 16–30 (2021)
20. Zimmermann, H.: OSI reference model-the ISO model of architecture for open systems interconnection. IEEE Trans. Commun. **28**(4), 425–432 (1980)
21. Preece, A., Harborne, D., Braines, D., Tomsett, R., Chakraborty, S.: Stakeholders in explainable AI. arXiv preprint arXiv:1810.00184. (2018)
22. Tomsett, R., Braines, D., Harborne, D., Preece, A., Chakraborty, S.: Interpretable to whom? A role-based model for analyzing interpretable machine learning systems. arXiv preprint arXiv:1806.07552. (2018)
23. Madaio, M., Egede, L., Subramonyam, H., Wortman Vaughan, J., Wallach, H.: Assessing the fairness of AI systems: AI practitioners' processes, challenges, and needs for support. Proc. ACM Hum.-Comput. Interact. **6**(CSCW1), 1–26 (2022)
24. Langer, M., Baum, K., Hartmann, K., Hessel, S., Speith, T., Wahl, J., Explainability auditing for intelligent systems: a rationale for multi-disciplinary perspectives. In: IEEE 29th International Requirements Engineering Conference Workshops (REW). IEEE **2021**, 164–168 (2021)
25. Shin, D.: Toward fair, accountable, and transparent algorithms: case studies on algorithm initiatives in Korea and China. Javnost Public. **26**(3), 274–290 (2019)
26. Saldanha, D.M.F., Dias, C.N., Guillaumon, S.: Transparency and accountability in digital public services: learning from the Brazilian cases. Gov. Inf. Q. **39**(2), 101680 (2022)
27. Bilan, Y., Mishchuk, H., Samoliuk, N., Mishchuk, V.: Gender discrimination and its links with compensations and benefits practices in enterprises. Entrep. Bus. Econ. Rev. **8**(3), 189–203 (2020)
28. Esses, V.M.: Prejudice and discrimination toward immigrants. Annu. Rev. Psychol. **72**, 503–531 (2021)
29. Yan, E., Lai, D.W., Lee, V.W., Bai, X., KLNg, H.: Abuse and discrimination experienced by older women in the era of representative COVID-19: a two-wave community survey in Hong Kong. Violence Against Women **28**(8), 1750–1772 (2022)
30. Wadsworth, C., Vera, F., Piech, C.: Achieving fairness through adversarial learning: an application to recidivism prediction. arXiv preprint arXiv:1807.00199. (2018)
31. Chowdhury, R., Mulani, N.: Auditing algorithms for bias. Harvard Bus. Rev. **24** (2018)
32. Manrai, A.K., Funke, B.H., Rehm, H.L., Olesen, M.S., Maron, B.A., Szolovits, P., et al.: Genetic misdiagnoses and the potential for health disparities. N. Engl. J. Med. **375**(7), 655–665 (2016)
33. Shankar, S., Halpern, Y., Breck, E., Atwood, J., Wilson, J., Sculley, D.: No classification without representation: assessing geodiversity issues in open data sets for the developing world. arXiv preprint arXiv:1711.08536. (2017)
34. Kodiyan, A.A.: An overview of ethical issues in using AI systems in hiring with a case study of Amazon's AI based hiring tool. Researchgate Preprint. pp. 1–19 (2019)
35. Ajunwa, I.: Beware of automated hiring. The New York Times. **8** (2019)
36. Baeza-Yates, R.: Bias on the web. Commun. ACM **61**(6), 54–61 (2018)
37. Akter, S., Dwivedi, Y.K., Biswas, K., Michael, K., Bandara, R.J., Sajib, S.: Addressing algorithmic bias in AI-driven customer management. J. Global Inf. Manag. (JGIM). **29**(6), 1–27 (2021)
38. Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., Galstyan, A.: A survey on bias and fairness in machine learning. ACM Comput. Surv. (CSUR). **54**(6), 1–35 (2021)
39. Friedman, B., Nissenbaum, H.: Bias in computer systems. In: Computer Ethics. Routledge. pp. 215–232 (2017)
40. Schwartz, R., Vassilev, A., Greene, K., Perine, L., Burt, A., Hall, P., et al.: Towards a Standard for Identifying and Managing Bias in Artificial Intelligence. Special Publication (NIST SP), National Institute of Standards and Technology. (2022)
41. Castelnovo, A., Crupi, R., Greco, G., Regoli, D., Penco, I.G., Cosentini, A.C.: A clarification of the nuances in the fairness metrics landscape. Sci. Rep. **12**(1), 1–21 (2022)
42. Hardt, M., Price, E., Srebro, N.: Equality of opportunity in supervised learning. Adv. Neural Inf. Process. Syst. **29** (2016)
43. Hinnefeld, J.H., Cooman, P., Mammo, N., Deese, R.: Evaluating fairness metrics in the presence of dataset bias. arXiv preprint arXiv:1809.09245. (2018)
44. Pandit, S., Gupta, S., et al.: A comparative study on distance measuring approaches for clustering. Int. J. Res. Comput. Sci. **2**(1), 29–31 (2011)

45. Agarwal, A., Agarwal, H., Agarwal, N.: Fairness Score and process standardization: framework for fairness certification in artificial intelligence systems. AI and Ethics. p. 1–13 (2022)
46. Shearer, C.: The CRISP-DM model: the new blueprint for data mining. J. Data Warehous. **5**(4), 13–22 (2000)
47. Fayyad, U., Piatetsky-Shapiro, G., Smyth, P.: The KDD process for extracting useful knowledge from volumes of data. Commun. ACM **39**(11), 27–34 (1996)
48. Azevedo, A., Santos, M.F.: KDD, SEMMA and CRISP-DM: a parallel overview. IADS-DM. (2008)
49. Shafique, U., Qaiser, H.: A comparative study of data mining process models (KDD, CRISP-DM and SEMMA). Int. J. Innov. Sci. Res. **12**(1), 217–222 (2014)
50. Amershi, S., Begel, A., Bird, C., DeLine, R., Gall, H., Kamar, E., et al.: Software engineering for machine learning: a case study. In: 2019 IEEE/ACM 41st International Conference on Software Engineering: Software Engineering in Practice (ICSE-SEIP). IEEE pp. 291–300 (2019)
51. De Silva, D., Alahakoon, D.: An artificial intelligence life cycle: from conception to production. Patterns **3**(6), 100489 (2022)
52. Wang, L., Liu, Z., Liu, A., Tao, F.: Artificial intelligence in product lifecycle management. Int. J. Adv. Manuf. Technol. **114**, 771–796 (2021)
53. Suresh, H., Guttag, J.: A framework for understanding sources of harm throughout the machine learning life cycle. In: Equity and Access in Algorithms, Mechanisms, and Optimization. pp. 1–9 (2021)
54. Fahse, T., Huber, V., van Giffen, B.: Managing bias in machine learning projects. In: Innovation Through Information Systems: Volume II: A Collection of Latest Research on Technology Issues. Springer. p. 94–109 (2021)
55. Bantilan, N.: Themis-ml: a fairness-aware machine learning interface for end-to-end discrimination discovery and mitigation. J. Technol. Hum. Serv. **36**(1), 15–30 (2018)
56. Ashmore, R., Calinescu, R., Paterson, C.: Assuring the machine learning lifecycle: desiderata, methods, and challenges. ACM Comput. Surv. (CSUR) **54**(5), 1–39 (2021)
57. Raji, I.D., Smart, A., White, R.N., Mitchell, M., Gebru, T., Hutchinson, B., et al.: Closing the AI accountability gap: defining an end-to-end framework for internal algorithmic auditing. In: Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency. pp. 33–44 (2020)
58. Mökander, J., Floridi, L.: Operationalising AI governance through ethics-based auditing: an industry case study. AI and Ethics pp. 1–18 (2022)
59. Klein, E.: Validation of a framework for bias identification and mitigation in algorithmic systems. Int. J. Adv. Softw. **14**(1 &2), 59–70 (2021)
60. Wang, H.E., Landers, M., Adams, R., Subbaswamy, A., Kharrazi, H., Gaskin, D.J., et al.: A bias evaluation checklist for predictive models and its pilot application for 30-day hospital readmission models. J. Am. Med. Inform. Assoc. **29**(8), 1323–1333 (2022)
61. Fabbrizzi, S., Papadopoulos, S., Ntoutsi, E., Kompatsiaris, I.: A survey on bias in visual datasets. Comput. Vis. Image Underst. **223**, 103552 (2022)
62. Madaio, M.A., Stark, L., Wortman Vaughan, J., Wallach, H.: Co-designing checklists to understand organizational challenges and opportunities around fairness in AI. In: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. pp. 1–14 (2020)
63. Richardson, B., Gilbert, J.E.: A framework for fairness: a systematic review of existing fair AI solutions. arXiv preprint arXiv:2112.05700. (2021)
64. Seedat, N., Imrie, F., van der Schaar, M.: DC-Check: A Data-Centric AI checklist to guide the development of reliable machine learning systems. arXiv preprint arXiv:2211.05764. (2022)
65. Ryan, M., Stahl, B.C.: Artificial intelligence ethics guidelines for developers and users: clarifying their content and normative implications. J. Inf. Commun. Ethics Soc. (2020)
66. Jobin, A., Ienca, M., Vayena, E.: The global landscape of AI ethics guidelines. Nat. Mach. Intell. **1**(9), 389–399 (2019)
67. HLEG, A.: Assessment list for trustworthy artificial intelligence (ALTAI) for self-assessment. High Level Expert Group on Artificial Intelligence B-1049 Brussels. (2020)
68. Kumar, A., Braud, T., Tarkoma, S., Hui, P.: Trustworthy AI in the age of pervasive computing and big data. In: 2020 IEEE International Conference on Pervasive Computing and Communications Workshops (PerCom Workshops). IEEE. pp. 1–6 (2020)
69. Bateni, A., Chan, M.C., Eitel-Porter, R.: AI fairness: from principles to practice. arXiv preprint arXiv:2207.09833. (2022)
70. Gupta, D., Krishnan, T.: Algorithmic bias: Why bother. California Manag. Rev. **63**(3) (2020)
71. Sorokin, A., Forsyth, D., Utility data annotation with amazon mechanical turk. In: IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops. IEEE **2008**, 1–8 (2008)
72. Northcutt, C.G., Athalye, A., Mueller, J.: Pervasive label errors in test sets destabilize machine learning benchmarks. arXiv preprint arXiv:2103.14749. (2021)
73. Buolamwini, J., Gebru, T.: Gender shades: intersectional accuracy disparities in commercial gender classification. In: Conference on Fairness, Accountability and Transparency. PMLR. pp. 77–91 (2018)
74. Fu, R., Huang, Y., Singh, P.V.: AI and algorithmic bias: source, detection, mitigation and implications. Detect. Mitigat. Implicat. (July 26, 2020). (2020)
75. Srinivasan, R., Chander, A.: Biases in AI systems. Commun. ACM **64**(8), 44–49 (2021)
76. Ayres, I.: Testing for discrimination and the problem of "included variable bias". Yale Law School Mimeo. (2010)
77. U S Bureau of Labor Statistics.: Atus Home. [Online; accessed 4-Feb-2023]. Available from: https://www.bls.gov/tus
78. Kamiran, F., Calders, T.: Data preprocessing techniques for classification without discrimination. Knowl. Inf. Syst. **33**(1), 1–33 (2012)
79. Calmon, F., Wei, D., Vinzamuri, B., Natesan Ramamurthy, K., Varshney, K.R.: Optimized pre-processing for discrimination prevention. Adv. Neural Inf. Process. Syst. **30** (2017)
80. Polli F.: The dark side of artificial intelligence
81. Whiteford, P.: Debt by design: The anatomy of a social policy fiasco-or was it something worse? Aust. J. Public Adm. **80**(2), 340–360 (2021)
82. Wakabayashi, D.: Self-driving Uber car kills pedestrian in Arizona, where robots roam. The New York Times. **19**(03) (2018)
83. Shah, S.: Amazon workers hospitalized after warehouse robot releases bear repellent
84. Siwicki, B.: How AI bias happens – and how to eliminate it
85. Angelino, E., Larus-Stone, N., Alabi, D., Seltzer, M., Rudin, C.: Learning certifiably optimal rule lists for categorical data. J. Mach. Learn. Res. **18**, 1–78 (2018)
86. Aggarwal, A., Lohia, P., Nagar, S., Dey, K., Saha, D.: Black box fairness testing of machine learning models. In: Proceedings of the 2019 27th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering. pp. 625–635 (2019)
87. Kazim, E., Koshiyama, A.S., Hilliard, A., Polle, R.: Systematizing audit in algorithmic recruitment. J. Intell. **9**(3), 46 (2021)

88. Landers, R.N., Behrend, T.S.: Auditing the AI auditors: a framework for evaluating fairness and bias in high stakes AI predictive models. Am. Psychol. (2022)
89. Dua, D., Graff, C.: UCI Machine Learning Repository. Available from: http://archive.ics.uci.edu/ml